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AI CAREERS FOR WOMEN

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PRESENTATION 2026

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Outline

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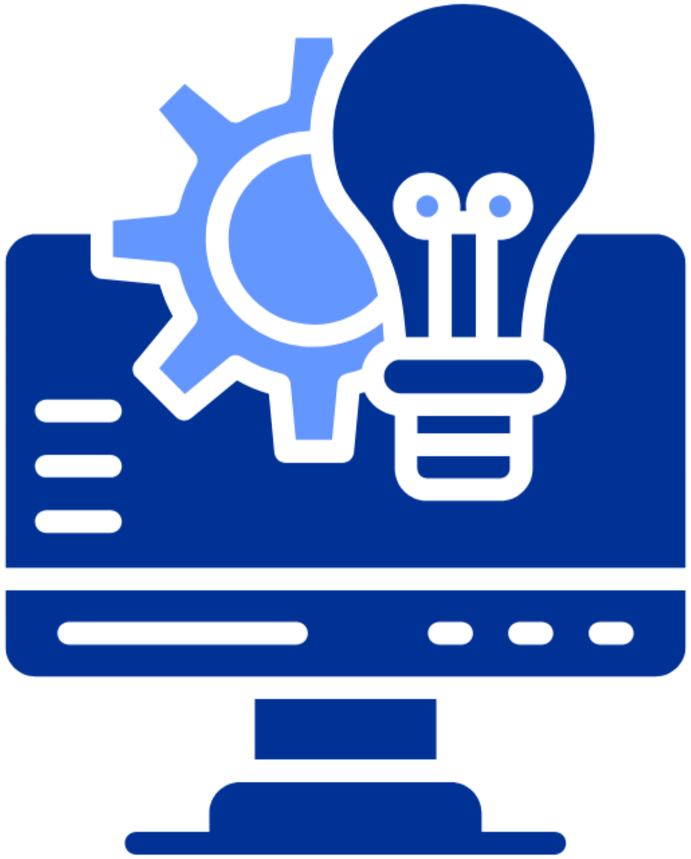


Abstract

- **Payroll Data Intelligence for SMEs**

Excel + AI-Powered Analytics

- **Project Objective:**

  Clean & analyze 5000-row payroll data

  . Generate business insights

- **Tools Used:**

  Microsoft Excel 365 + Copilot

- **Key Deliverables:**

  High data accuracy after cleaning

  Interactive Excel dashboard (charts & insights)

  TDS & PF compliance validation

  Department-wise salary analysis

- **Unique Value:**

  Zero-cost analytics solution

  Converts manual payroll data into validated dashboards using Excel + AI

- **Business Impact**:

  Reduces payroll errors and compliance risks

  Provides practical, ready-to-use business insights

Problem Statement

- SMEs rely heavily on manual Excel-based payroll systems
- High risk of errors in salary calculations and deductions
- **Key Challenges**:
  - Incorrect Gross and Net Salary calculations
  - Duplicate records and missing values
  - Lack of automated validation framework
- **Business Impact:**
  - Compliance risks (TDS/PF errors)
  - Financial inaccuracies and audit issues
  - Inefficient reporting and decision-making
- **Limitations of Existing Approach:**
  - Manual validation is time-consuming
  - No structured system for error detection
- **Scope**:
  - Focus on payroll data cleaning, validation, and analysis using Excel + AI tools

Proposed Solution

- Developed a structured Excel-based payroll validation system integrated with AI tools

- **Approach**
  Data Cleaning: Removed duplicates, handled missing values, standardized formats
  Data Validation: Applied logical rules for salary calculations
  Data Analysis: Used PivotTables and charts for insights

- **AI Integration**:
  ChatGPT used for validation logic and insights
  Perplexity used for research and formula support

- **Key Benefits:**
  Improves payroll accuracy and consistency
  Reduces manual effort and processing time
  Provides business-ready dashboards

- **Innovation**:
  Combines traditional Excel with AI for intelligent data analysis
  Cost-effective solution for SMEs

System Architecture

Data Input / Source

- Data collected from Excel payroll datasets

Includes Employee ID, Department, Designation, Salary details, Pay Month

- Stored in Raw Data sheet
- Cleaned and maintained in Clean Data sheet for analysis

Processing / Analysis

- Cleaned data using Excel (removed duplicates, handled missing values, standardized formats)
- Created calculations: Gross Salary and Net Salary
- Used Pivot Tables & Charts for analysis
- Analyzed monthly trends, department-wise, and designation-wise salary

AI / Tool Integration

- Used Microsoft Excel for data processing and dashboard creation
- Applied Pivot Tables, Pivot Charts & Conditional Formatting
- Utilized Microsoft Copilot for formula and design guidance
- Fully Excel-based solution (no external tools required)

Output / Dashboard

- Final output is a professional Excel payroll dashboard
- Includes KPIs (Total Salary, Net Salary, Employee Count)
- Features charts for monthly, department, and designation analysis
- Interactive, visually appealing, and supports decision-making

Live Demo of Project

Here's the list of some demo screenshots / features of the project:

Dashboard Overview

- Shows the main dashboard layout
- Displays KPIs like total salary, net salary, and employee count
- Includes multiple charts for quick insights
- Designed with a clean and corporate UI

Department Analysis

- Displays salary distribution across departments
- Uses bar/pie charts for comparison
- Helps identify:

Highest paying department

- Department-wise cost analysis

Monthly Analysis

- Shows salary trends over months
- Uses line/column charts

Helps track:

- Payroll growth

Monthly variations

- Business performance trends

[https://your-project-demo-link.com]

[Insert your project demo video.
You can embed a video your live project.]

Conclusion

Successfully cleaned and validated a 5000-row payroll dataset

- Achieved high data accuracy and consistency

- **Key Results**:

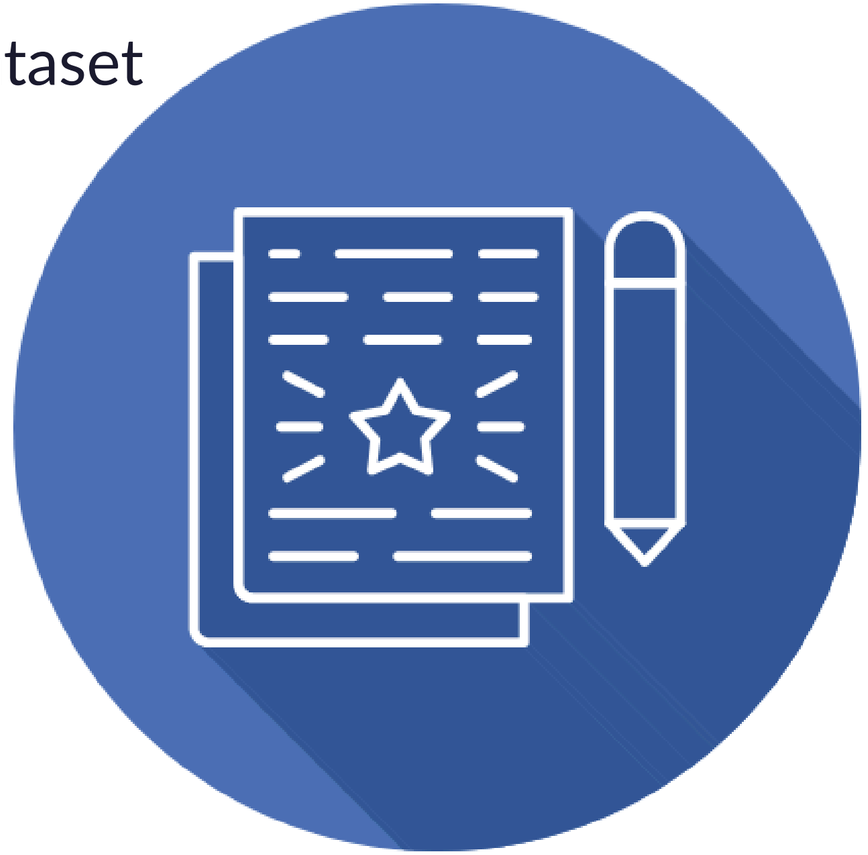
– Verified salary relationships (Gross and Net Salary)
– Identified and removed inconsistencies
– Generated actionable business insights

- **Impact**:

– Reduced analysis time significantly
– Improved payroll reliability and compliance

- **Value Addition**:

– Demonstrates practical use of Excel + AI
– Provides scalable solution for SME payroll systems



Future Scope

Integration with real-time payroll systems

- Development of automated validation tools

- **Advanced Enhancements:**
  – Use of Power BI for advanced dashboards
  – Machine learning for anomaly detection

- **Business Expansion:**
  – Scalable solution for large datasets
  – Integration with HRMS and ERP systems

- **Compliance Improvement:**
  – Automated TDS/PF calculation updates
  – Real-time audit and reporting system

Thank You

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